Dependence and Correlation

Computing Fundamentals for Social Sciences

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Outline

Why Study Dependence?

- Understanding relationships between variables is fundamental in statistics
- Dependence tells us whether and how variables move together
- Applications: predicting one variable from another, identifying risk factors, exploring patterns

Types of Dependence

- **Mean dependence**: the mean of one variable depends on the level of another
- **Linear dependence**: changes in one variable tend to accompany proportional changes in another
- **Correlation**: standardized measure of linear association

Mean Dependence: Concept

- A variable Y is mean-dependent on X if the expected value of Y changes with X
- Expressed as $E[Y|X = x]$ varies with x
- Focuses on conditional averages rather than full distribution

Example in R: Mean Dependence

```r
# Average Score by Gender
aggregate(Score ~ Gender, data = data, FUN = mean)
```

- Shows how mean Score changes across categories of Gender

Visualizing Mean Dependence

```
ggplot(data, aes(x = Gender, y = Score)) +
  geom_boxplot(fill = "lightblue") +
  stat_summary(fun = mean, geom = "point", color = "red",
     size = 3) +
  labs(title = "Score by Gender with Mean Points")
```

Linear Dependence: Concept

- Linear dependence means one variable increases (or decreases) roughly proportionally with another
- Can be visualized as points around a straight line in a scatterplot
- Not all dependence is linear: e.g., quadratic relationships exist

Scatterplot Example

```
ggplot(data, aes(x = Height, y = Score)) +
  geom_point(color = "blue") +
  labs(title = "Scatterplot of Height vs Score")
```

Adding a Regression Line

```
ggplot(data, aes(x = Height, y = Score)) +
  geom_point(color = "blue") +
  geom_smooth(method = "lm", color = "red") +
  labs(title = "Height vs Score with Regression Line")
```

Interpretation of Linear Dependence

- Slope indicates average change in Y for one unit change in X
- If slope ≈ 0, weak or no linear dependence
- Scatterplot shape helps detect non-linear dependence

Correlation: Definition

- Measures the strength and direction of linear association between two variables
- Pearson correlation coefficient r:

$$r = \frac{\mathrm{Cov}(X, Y)}{SD(X)SD(Y)}$$

- $r \in [-1, 1]$, where:
  - $r \approx 1$: strong positive linear association
  - $r \approx -1$: strong negative linear association
  - $r \approx 0$: weak/no linear association

Correlation in R

```
# Pearson correlation between Height and Score
cor(data$Height, data$Score)
```

Correlation with Scatterplot

```
ggplot(data, aes(x = Height, y = Score)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red") +
  labs(title = paste("Correlation:", round(cor(data$Height,
     data$Score),2)))
```

Exercise 1

- Compute the mean Score for each Gender
- Interpret mean dependence

Exercise 2

- Plot scatterplot of Height vs Score
- Add regression line
- Comment on linear dependence

Exercise 3

- Compute correlation between Height and Score
- Interpret its magnitude and sign

Exercise Solutions

```r
# Exercise 1
aggregate(Score ~ Gender, data = data, FUN = mean)

# Exercise 2
ggplot(data, aes(x = Height, y = Score)) +
  geom_point() +
  geom_smooth(method = "lm", color = "red")

# Exercise 3
cor(data$Height, data$Score)
```

- Interpretation: positive correlation → taller individuals tend to score higher (example)
- Mean Score differs by Gender → mean dependence present

Summary

- Dependence helps understand relationships between variables
- Mean dependence captures conditional averages
- Linear dependence focuses on proportional changes
- Correlation quantifies linear association
- Visual tools (scatterplots, boxplots) complement numeric measures